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The Mystery of the White Dwarfs

in which Eddington and Chandrasekhar do battle
over the deaths of massive stars;
must they shrink when they die,
creating black holes?
or will quantum mechanics save them?

The year was 1928; the place, southeast India, the city of Madras on the Bay of Bengal. There, at Presidency College, a seventeen-year-old Indian boy named Subrahmanyan Chandrasekhar was immersed in the study of physics, chemistry, and mathematics. Chandrasekhar was tall and handsome, with regal bearing and pride in his academic achievements. He had recently read Arnold Sommerfeld's classic textbook *Atomic Structure and Spectral Lines* and was now overjoyed that Sommerfeld, one of the world's great theoretical physicists, had come from his home in Munich to visit Madras.

Eager for personal contact, Chandrasekhar went to Sommerfeld's hotel room and asked for an interview. Sommerfeld granted an appointment for several days hence.

On the day of his appointment Chandrasekhar, filled with pride and confidence in his mastery of modern physics, walked up to Sommerfeld's hotel room and knocked on the door. Sommerfeld greeted him politely, inquired about his studies, then deflated him. "The physics you have been studying is a thing of the past. Physics has all changed in the five years since my textbook was written," he explained. He went on to describe a revolution in physicists' understanding of the laws that govern the realm of